



NORSK DATA

PRODUCT INFORMATION

Product name

TCP/IP Basic Module/III

Product number

211327D

DESCRIPTION OF PRODUCT

To allow use of TCP/IP protocols by applications (FTP, TELNET) in a SINTRAN environment on a Ethernet local area network.

SIBAS backend is available over the TCP/IP Basic Module/III.

NFS support may be selected when installing this product.

This product supports startup and error reporting to a NCS network server. The Internet address of the NCS server must be put into the AIP-HOSTS:SYMB file with the name AUDITSERVER to achieve this. The product will also answer alive check polling from the NCS server.

REASON(S) FOR NEW PRODUCT OR VERSION

Version D00

Improved performance in the Tcp-Ip domino processes

Improved error reporting.

Meeting requirements from other products.

Makes it possible to display more informations from the TCP-Monitor

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1 PREREQUISITES

The following things are required in order to use this product.

1.1 HARDWARE PREREQUISITES

The following hardware must be present for this product to be used:

CPU type (any of the following):

- 5000

Microprogram Version for 5000

ND Product Number	System type	Microprogram Version
211272 C	ND-5200	11529
211273 C	ND-5400	11629
211274 C	ND-5500	11729
211275 C	ND-5700	11829
211276 C	ND-5800	11929
211847 A	ND-5830	12009
211847 A	ND-5850	12009

(or later
versions ...)

Other hardware:

Product name

Product number

Ethernet III controller

ND 110513

1.2 SOFTWARE PREREQUISITES

The following software must be installed on the system for this product to be used:

Operating system:

Name

Version/Revision

SINTRAN-III/VSX
Patch-file level >=4300
Or
SINTRAN-III/VSX
Patch-File level >=4200

L Work-Mode 7

.M Work-Mode 6

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Other software:

Product nameProduct number

BACKUP-SYSTEM version >=I

210337

USER ENVIRONMENT version >=D03

210518

1.3 OTHER PREREQUISITES

The following are necessary for this product to be used.

Minimum Permanent Mass Storage:

User	Space (pages)	Number of files
SYSTEM	120	4
TCP-IP	400	13
UTILITY	200	1

	Minimum	Maximum
Number of segments (ND-100)	2	
Number of RT descriptions	2	

Space required on segment files: 72 pages

Requirements regarding MTAD resources per controller (one for each new connection used):

Max number of MTAD datafields: 124

Number of background programs: 124

Requirements regarding Nucleus resources for the first controller :

Message buffer space for system processes in pages: 16

Number of descriptors for all system processes: 32

Requirements regarding nucleus resources for each new controller:

Message buffer space for system processes in pages: 1

Number of descriptors for all system processes: 1

The maximum number of MTAD datafields (MTMAX) is changed with the SINTRAN-SERVICE-PROGRAM, the number of background programs and nucleus resources with the S3-CONFIG program.

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2 DEPENDENCIES

In addition to the above listed hardware and/or software prerequisites, the following dependencies exist between this and other products. The hardware and software listed here may, for example, be necessary to use the product's full functionality.

2.1 HARDWARE DEPENDENCIES

There are dependency relationships between this product and the following hardware:

None

2.2 SOFTWARE DEPENDENCIES

There are dependency relationships between this product and the following software:

Product nameProduct number

OWS Access Server for ND-5000

ND-211325

Nature of dependency

The installation of the OWS Access Server for ND-5000 depends on this product to be installed.

SIBAS/R Backend for ND-5000

ND-211404

Nature of dependency

This product must be installed if SIBAS Backend shall be used with TCP/IP communication.

2.3 OTHER DEPENDENCIES

In addition to the hardware and software dependency relationships listed above, you should be aware of the following dependencies:

None.

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211327D**3 DOCUMENTATION**Included (distributed free with product)TitleNumber

OpenLAN Network Supervisor Guide

830107.03 EN

Optional (must be purchased)TitleNumber

COSMOS TELNET/FTP User Guide

860284.02 EN

4 MAINTENANCE SOFTWARE

Included in the TCP/IP Basic Module/III, on diskette no. 2, is the TCPIP-MONITOR, intended for use by ND technical support.

Other maintenance tools are the Nucleus and Domino maintenance kits. They are independant products.

Product NameProduct Number

Nucleus Maintenance Kit version C

ND-211321

Domino Maintenance Kit version C

ND-211322

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5 INSTALLATION PROCEDURE

5.1 SELECTING INTERNET ADDRESS

- On request from the installation program you will have to assign an Internet address for each controller.

If your net is isolated from other nets outside your company, you may select the address freely, i.e. using the default values given by the installation program.

If your net connects to other nets, free selection of addresses may accidentally give the same address to two installations which will create severe problems.

Selecting proper Internet addresses are further described in the OpenLAN Network Supervisor guide. However, the address for application of an official Internet address range is:

DDN Network Information Center
SRI International
Room EJ217
333 Ravenswood Avenue
Menlo Park, CA 94025

5.2 AUTOMATIC INSTALLATION

- Check that the RT-program PROMAN is running by giving the command:
@LIST-RT-DESCRIPTION PROMAN

If the program is not running, the first line will be:

PASSIVE

Start PROMAN by giving the command @START-SERVERS

If PROMAN starts without giving any error messages, the installation can continue.

If PROMAN fails to start, the command @START-SERVERS should be included in the LOAD-MODE:MODE or LOAD-SYSTEM:BATC as described in chapter 5.3, and the system should be restarted before installation is continued.

The installation will fail if PROMAN is passive.

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- If you want to keep the previous version of this product you must save the image file for that version: Log in as user UTILITY and rename the '(UTILITY)PMA-ETH3-TCP*:IMAG' file(s) to e.g. ':SAVE'.
- Log in as user SYSTEM
- Insert the floppy 211327D-XX-01D in the floppy-drive
@ENTER-DIRECTORY,211327,F-D-1,(0/1)
- Run the installation program:
@(211327:FLOPPY-USER)IN-TCP-BA
- The installation program will ask you whether or not you require NFS support. You should answer NO to this question unless you are actually going to use NFS, otherwise a lot of TCP/IP resources will be wasted.
- The installation program will prompt for the number of Telnet server connections to be configured. The maximum number of telnet server connections is 124

5.3 COMPLETING THE INSTALLATION

When the installation program has been run :

- Update the AIP-HOSTS:SYMB file with the local internet address of the host name. See the documentation for more information.
- Update the mode file that is run after a cold start (usually called HENT-MODE:MODE) by including the following line:

@MODE (TCP-IP)TCP-BA-LOAD-D:MODE,...

If you had a previous version of this product, remove the old line loading that version.

- Insert the line: @START-SERVERS

if you have not already done it, in the LOAD-MODE:MODE or LOAD-SYSTEM:BATC file after starting the XMSG (and after the command @START-TADADM which usually lies in the XMSG startup file)

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- If you are re-installing this product you have to do a warm start afterwards to make the re-installed product functions properly with respect to any changes to its configuration.
For example, in terms of Telnet server connections.
- To reboot the domino controller(s) later, run the mode job:
@MODE (TCP-IP)TCP-BA-REBO:M,,
- To stop the domino controller(s) later, run the mode job:
@MODE (TCP-IP)TCP-BA-STOP:M,,

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6 MODIFICATIONS

- Some error reports from the Eth-III controller software has been changed: eg. the "Duplicated IP addresses" report now gives both the IP and Ethernet addresses of the offending system. The ARP command of the TcpIp-Monitor enables you to find supposedly correct Ethernet address for that IP address.

7 ERRORS CORRECTED

- TCPD and waste of Nucleus resources
The previous versions of this product did not release the Nucleus resources involved in error reporting from the Eth-III controller software to the AuditServer . The nucleus resources will be wasted This has been corrected in TCPD rt-program.
- Telnet server and excessive error reported
In the previous versions of this product, the Telnet server would sometimes continue to poll connections that was disconnecting, resulting in excessive error reports. Error of type : "illegal socket ID etc ...
This has been corrected
- UDP-broadcasts thrown away
The previous versions of this product did not handle UDP-broadcasts correctly, especially incoming UDP-broadcasts were thrown away.
- ARP-entries are kept for at least half an hour
The ARP-entries will be kept for at least half an hour while in the previous version , they were not kept for more than 3 minutes which causes a bad performance.
- Mbufs used were not released
The Mbufs used by the RAW protocol were not released when they were no longer needed.
- Nucleus resources not released
The Nucleus resources used by the controller when an application used SLib to send data with the RAW protocol, were not released when finished.
- Mixing TCP-port and UDP-port
In previous versions of this product, datagrams sent to local UDP-port number X would end up at local TCP-port number X if

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this did exist and had been created before the equally numbered UDP-port.

- The TCP-IP data checksum was not enabled.
The data checksum was not performed and bad data were accepted
- The heartbeat-check mechanism did not work

The heartbeat-check mechanism used to check if the Eth-III controller is still alive did not work and the Eth-III controller will terminate. due to lack of activity.

The reason was actually that the Eth-III controller was active on a wrong level and the updating of the heartbeat was NOT done.

- Better performance
It is observed that the performance for the processes using TCP-IP is much better.

8 ERROR KNOWN BUT NOT CORRECTED

- None